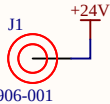
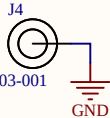


Inverting Buck-Boost

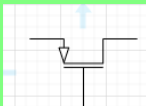


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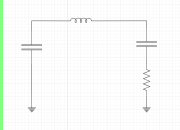
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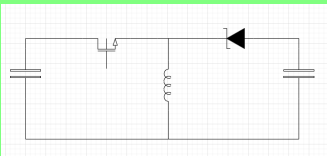


UVLO/OVLO

Input filter



Power stage



PWM

CSP

CSN

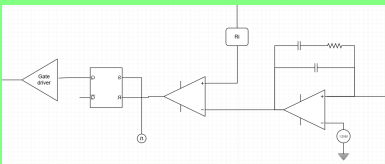
CSP

CSN

Current sense

CS

Control circuit

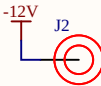


PWM

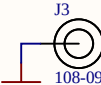
OCF

RAMP

FB

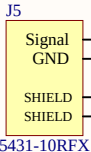
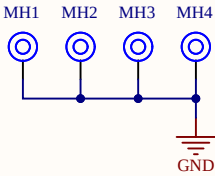
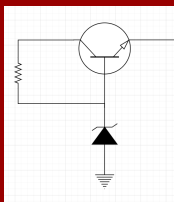


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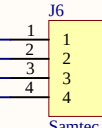


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Linear regulators



031-5431-10RFX



Samtec TS-104-T-AA

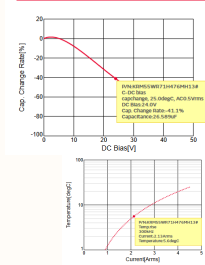
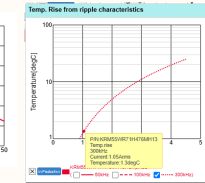
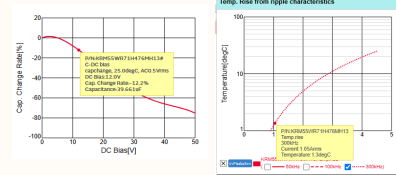
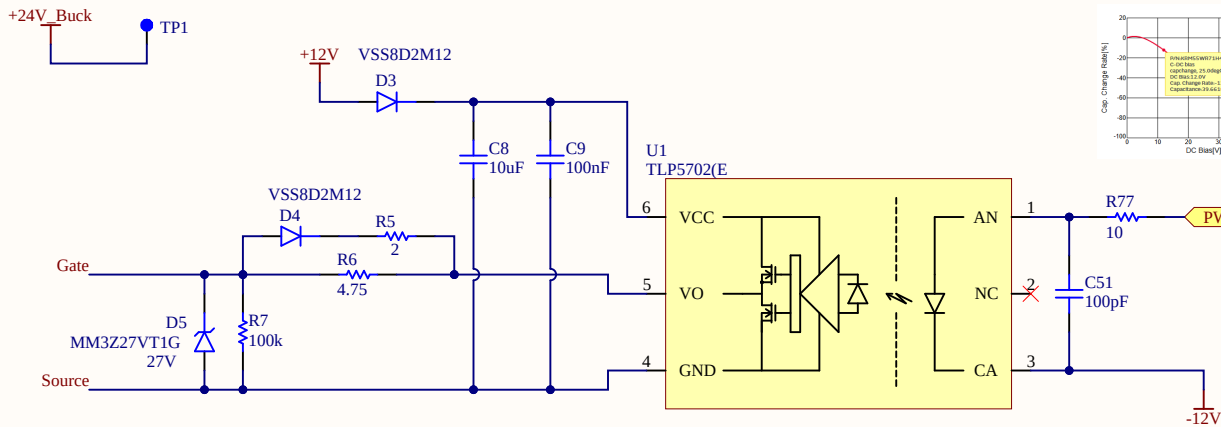
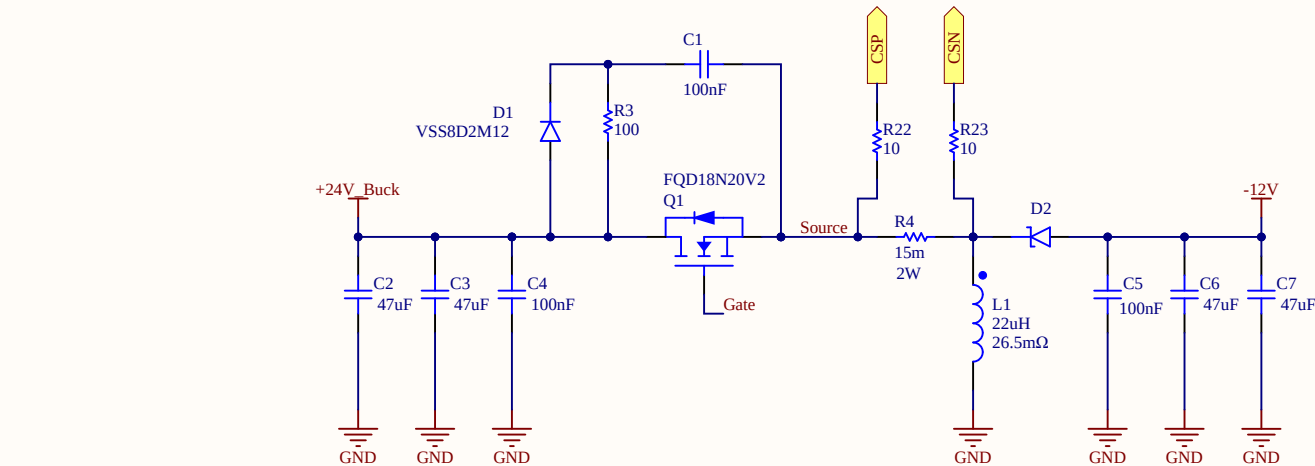
J6 is a 4-pin header that selects the op-amp negative input source: either the injected sawtooth signal (to subtract it from the compensator output) or -12 V to disable this feature.

J5 is a female BNC connector used to inject an external compensating ramp (sawtooth) if needed



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		Driss

Power stage



ANALYSIS

Vin=24V | Vout=-12V Iout=3A | fs=300kHz | D=0.333 | eff=88%

INDUCTOR — SRP1245A-220M (22uH)

$I_{L_avg} = I_{in} + I_{out} = 1.705 + 3 = 4.705A$
 $\Delta I_L = V_{in} \times D / (f_s \times L) = 1.21A$
 $I_{L_peak} = I_{L_avg} + \Delta I_L / 2 = 5.31A$
 $I_{L_ripple_ratio} = 1.21 / 4.705 = 25.7\%$
 $L_{crit} = (D')^2 \times R / 2f_s = 2.97uH$
CCM confirmed: $L = 22uH \gg L_{crit} = 2.97uH$ ✓

MOSFET — FQD18N20V2 (200 V, 15 A, 140 mΩ)

$V_{ds_stress} = V_{in} + V_{out} = 24 + 12 = 36V$
 $V_{ds_rated} = 36 \times 1.2 = 43.2V \rightarrow 600V$ ✓ (20% derating)
 $I_{d_rated} \geq I_{L_peak} = 5.31A \rightarrow 15A$ ✓
 $I_{D_rms} = I_{L_avg} \times \sqrt{D} = 2.72A$
 $P_{cond} = I_{D_rms}^2 \times R_{ds(on)} \times 1.3 = 2.72^2 \times 0.637 = 4.71W$ (30% derating)
 $P_{sw} = 0.5 \times V_{ds} \times I_{L_peak} \times (t_r + t_f) \times f_s = 0.5 \times 36 \times 5.31 \times 16n \times 300k = 0.46W$
 $P_{total} = 4.71 + 0.57 = 5.12W$

DIODE — STPS10L60S (60V, 10A, Vf=0.4V, DPAK)

$V_r_stress = V_{in} + V_{out} = 36V$
 $V_r_rated = 36 \times 1.2 = 43.2V \rightarrow 60V$ ✓
 $I_{D_avg} = I_{out} = 3A$
 $I_{D_peak} = I_{L_peak} = 5.31A$
 $P_{diode} = V_f \times I_{out} \times (1-D) = 0.4 \times 3 \times 0.667 = 0.8W$

OUTPUT CAPACITOR — KRM55WR71H476MH13K ×2 (47uF,50V,X7R)

$C_{min} = I_{out} \times D / (f_s \times \Delta V_{out}) = 27.7uF$
 $C_{nominal} = 2 \times 47 = 94uF$
 $C_{derated} @ 12V = 2 \times 39.6 = 79.2uF > 27.7uF$ ✓
 $\Delta V_{out_target} = 1\% \times 12 = 120mV$
 $ESR_{max} = 0.5 \times \Delta V_{out} / \Delta I_L = 49.6m\Omega$
(the ESR can contribute max by a 50% to the total allowed ripple)
 $ESR_{eff} = 4m\Omega / 2 = 2m\Omega \ll 49.6m\Omega$ ✓
 $I_{C_rms} = I_{out} \times \sqrt{D(D')} = 2.12A$
 $I_{C_rms/cap} = 2.12 / 2 = 1.06A$ ✓

INPUT CAPACITOR — KRM55WR71H476MH13K ×2 (47uF,50V,X7R)

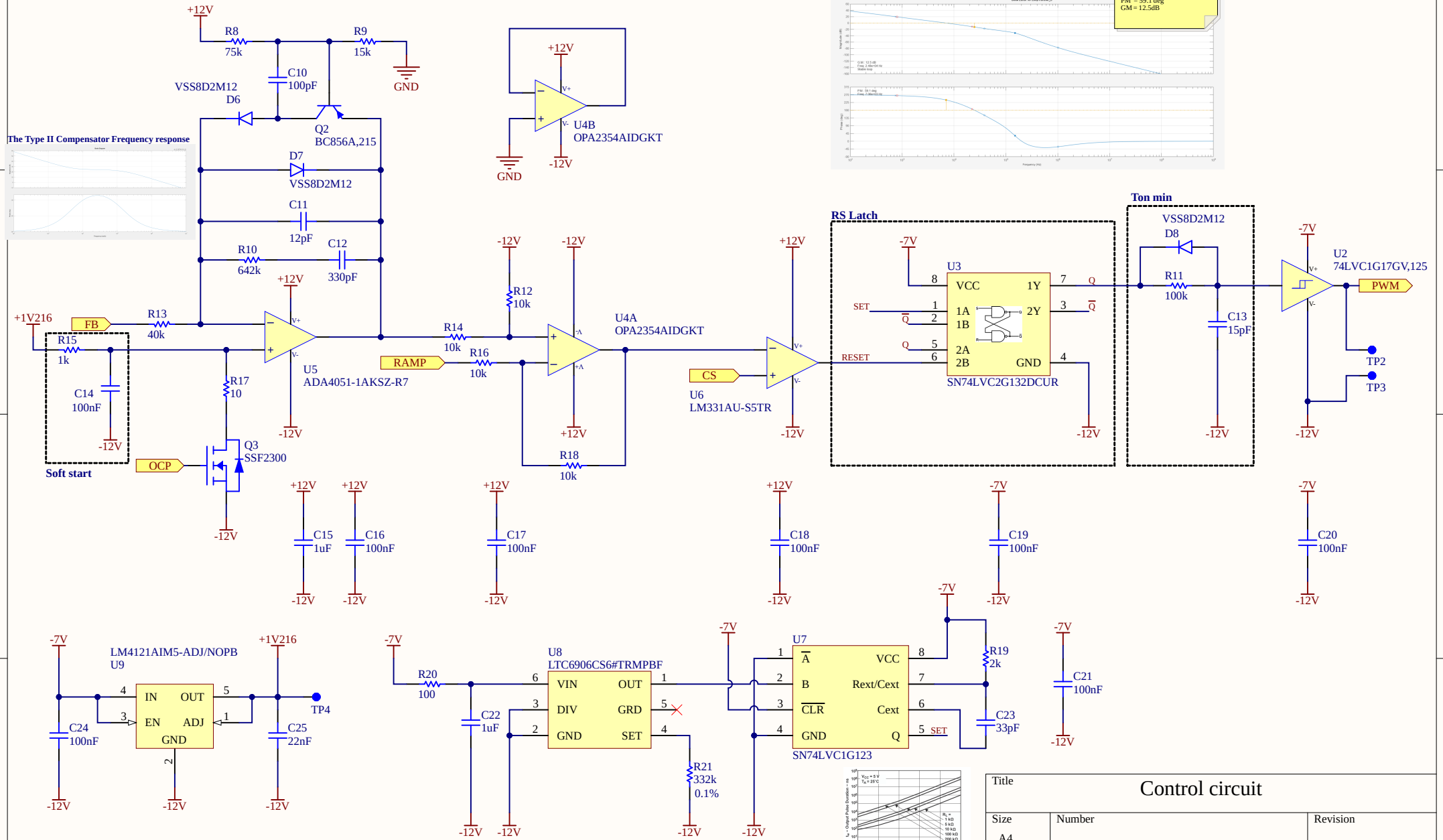
$C_{min} = I_{out} \times D / (f_s \times \Delta V_{in} \times D') = 20.8uF$
 $C_{nominal} = 2 \times 47 = 94uF$
 $C_{derated} @ 24V = 2 \times 26.6uF = 53.2uF > 20.8uF$ ✓
 $\Delta V_{in_target} = 1\% \times 24 = 240mV$
 $ESR_{max} = 0.5 \times \Delta V_{in} / I_{L_peak} = 22.6m\Omega$
 $I_{Cin_rms} = I_L \times \sqrt{D \times D'} = 2.22A$ ✓
 $V_{rated} = 32 \times 1.5 = 48V \rightarrow 50V$ ✓
 $C_{in} \text{ impedance } Z_{cin} = 1 / (2\pi \times f_s \times C_{in}) + ESR_{cin} = 0.413\Omega$
 $C_{damp} \text{ impedance } Z_{cdamp} = 1 / (2\pi \times f_s \times C_{damp}) + ESR_{damp} = 0.013\Omega$
 $IRMS \text{ Total Input Caps } (C_{in} + C_{damp}) = I_L \times \sqrt{D \times D'} = 2.22A$
 $IRMS \text{ } C_{damp} = Z_{cdamp} / (Z_{cdamp} + Z_{cin}) = 2.16A$ ✓

POWER BUDGET

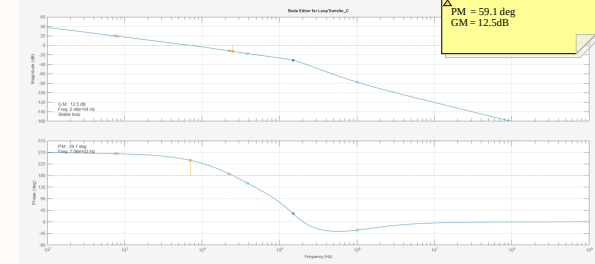
$P_{mosfet} = 5.17W$
 $P_{diode} = 0.80W$
 $P_{total} = 6.08W$
 $\eta_{estimated} = P_{out} / (P_{out} + P_{loss}) = 36 / (36 + 6.08) = 85.6\%$

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Power stage		
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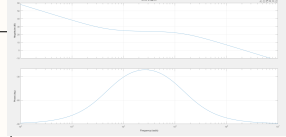
Control circuit



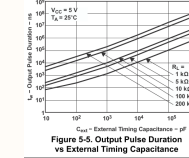
Closed Loop Gain



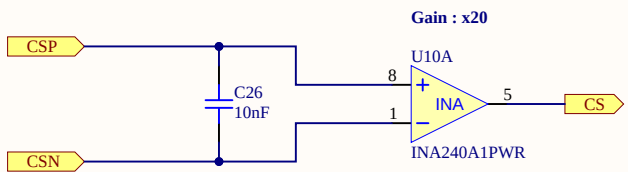
The Type II Compensator Frequency response



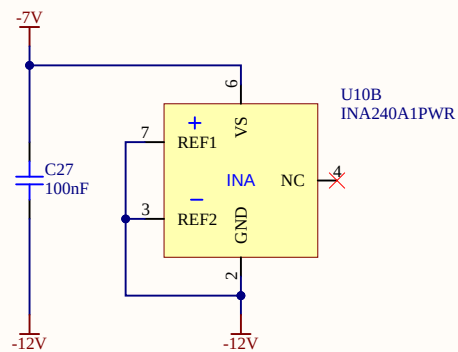
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Current sense

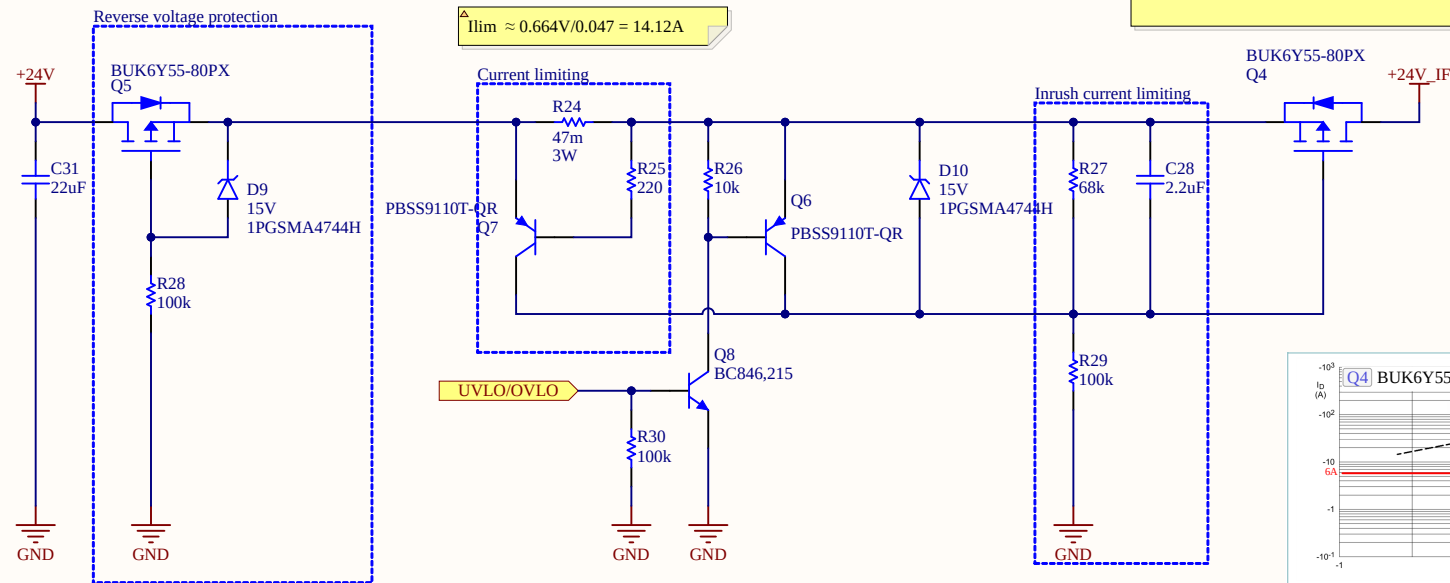


Total current gain = 0.015 x 20 = 0.3



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Input switch

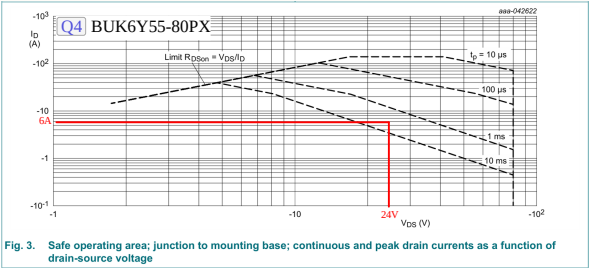


$I_{lim} \approx 0.664V / 0.047 = 14.12A$

STATIC POWER DISSIPATION for Q4

$P = I_{in}^2 \times R_{DS(on)}$
 $= (1.705 A)^2 \times 0.055 \Omega$
 $= 0.160 W$

Maximum static power dissipation: 160 mW
Estimated junction-to-case temperature rise: 0.12 °C
(RθJ-MB = 1 °C/W)



$C_{load} = C_f + C_{damp} + C_{in}$
 $= 5.45u + 53.3u + 220u = 278.75u \approx 300u$
 $\Delta t = C_{load} \times V_{in} / I_{inrush_max}$
 $\Delta t = 300uF \times 24V / 6A = 1.2ms$ (The MOSFET Q4 can withstand pulses longer than 1 ms. The SOA analysis is a conservative method for verifying whether the transistor can support the inrush current while operating in the linear region. In this analysis, the current is assumed to be a square wave and the VDS is assumed to remain constant at 24 V throughout the entire 1.2 ms inrush period. Based on these conservative assumptions, the MOSFET can be considered suitable for this application.)
based on LTSpice simulation a cap value of 2.2uF C28
in combination with 100K R29 we can achieve approx $I_{inr} = 5.8A$ within 800us which is sufficient and Q4 can withstand
 $V_{gate(steady\ state)} = V_{in} \times R_{29} / (R_{27} + R_{29}) = 24 \times 100k / (68k + 100k) = 24 \times 0.595 = 14.3V$
 $V_{GS,ss} = V_{gate,ss} - V_{in} = 14.3 - 24 = 9.7V$ more than Vgs threshold and less than Vgs max of

Input switch		
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INPUT EMI FILTER : Design Notes

Standard: EN 55032:2015 Class B — Conducted Emissions

Limit @ 300kHz = 56 dBμV (quasi-peak)

Frequency range: 150kHz – 30MHz

SPECIFICATIONS:

Vin = 24V | Vout = -12V | Iout = 3A | fs = 300kHz | D = 0.333 | Cin = 53.2uF (derated @24V)

STEP 1 : NOISE LEVEL ESTIMATE

First harmonic amplitude:

$$I_1 = (2 \times I_L \times \sin(\pi \times D)) / \pi = (2 \times 4.705 \times \sin(\pi \times 0.333)) / \pi = 2.60A$$

Noise voltage across Cin at fs:

$$V_{noise} = I_1 / (2\pi \times fs \times C_{in}) = 2.60 / (2\pi \times 300k \times 53.2u) = 25.9mV = 25.9 \times 10^3 \mu V$$

Estimated noise level:

$$|Noise|_{dB\mu V} = 20 \log(25.9 \times 10^3) = 88.3 \text{ dB}\mu V$$

Required attenuation:

$$Att = |Noise|_{dB\mu V} - V_{max} = 88.3 - 56 = 32.3 \text{ dB}$$

Adding 10dB safety margin:

$$Att_{target} = 32.3 + 10 = 42.3 \text{ dB}$$

STEP 2 : INDUCTOR SELECTION

Lf = 10uH → Selected: 10uH, 12A, SMD

STEP 3 : CAPACITOR SELECTION

Resonance one decade below fs:

$$C_f = C_{in} / (C_{in} \times L_f \times (2\pi \times fs / 10)^2 - 1) = 53.2u / (53.2u \times 10u \times (2\pi \times 30k)^2 - 1) = 3\mu F$$

Required attenuation:

$$C_f = 1 / L_f \times (10^{\wedge}(Att/40) / (2\pi \times fs))^2 = 5\mu F$$

Select higher value: Cf = max(5μF, 3μF) = 5μF → 10μF selected derated to 5.45μF @24V

NOTE: Cf is a SEPARATE dedicated filter capacitor distinct from Cin.

$$f_{c_filter} = 1 / (2\pi \times \sqrt{L_f \times C_f}) = 1 / (2\pi \times \sqrt{10u \times 5.45\mu F}) = 21.6kHz$$

Attenuation verification:

$$A = 40 \times \log(fs / f_{c_filter}) = 40 \times \log(300k / 21.6k) = 45.7dB \gg 42.3dB \checkmark$$

STEP 4 : DAMPING CAPACITOR

$$C_d \geq 4 \times C_{in} = 4 \times 53.2u = 212.8uF \rightarrow 220uF \text{ selected}$$

Required ESRd:

$$ESR_d \leq \sqrt{L_f / C_{in}} = \sqrt{10u / 53.2u} = 0.434\Omega$$

Damping resistor power:

$$P_{R_damp} \approx 0.5 \times \Delta I_{in}^2 \times R_{damp} = 0.5 \times (1.21)^2 \times 1.5 = 0.8W$$

Selected Cd: 220uF 35V polymer electrolytic ESR_polymer ≈ 11mΩ < 434mΩ ✓

NOTE: Polymer ESR too low, added series resistor Rd = 0.4Ω in series with Cd to meet ESRd target

Voltage rating: Vin_max × 1.25 = 32 × 1.25 = 40V → 50V ✓

$$C_{in} \text{ impedance } Z_{cin} = 1 / (2\pi \times fs \times C_{in}) + ESR_{cin} = 0.413\Omega$$

$$C_{damp} \text{ impedance } Z_{cdamp} = 1 / (2\pi \times fs \times C_{damp}) + ESR_{damp} = 0.013\Omega$$

$$IRMS \text{ Total Input Caps } (C_{in} + C_{damp}) = I_L \times \sqrt{D \times D'} = 2.22A$$

$$IRMS \text{ Cdamp (current divider)} = Z_{cin} / (Z_{cdamp} + Z_{cin}) = 0.062A \checkmark$$

STEP 4 : MIDDLEBROOK CRITERION

$$Z_{in_conv} = V_{in}^2 \times \eta / P_{out} = 576 \times 0.88 / 36 = 14.1\Omega$$

Full frequency spectrum verified using matlab: see Plot

EMC STANDARD : EN 55032:2015

EN 55032 "Electromagnetic Compatibility of Multimedia

Equipment". It defines conducted emission limits on the power port for Class B equipment (residential environment):

150kHz – 500kHz : 56–66 dBμV (quasi-peak)

46–56 dBμV (average)

500kHz – 30MHz : 56 dBμV (quasi-peak)

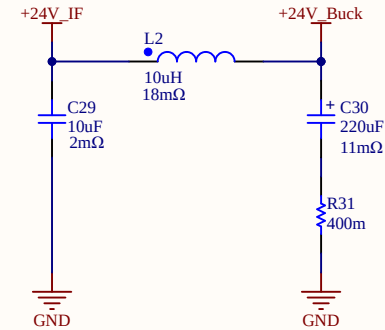
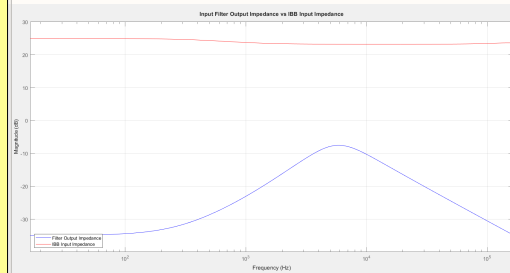
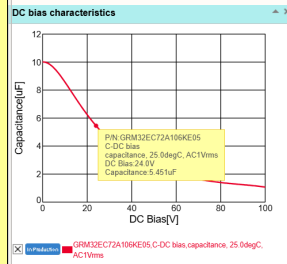
46 dBμV (average)

Worst case for this design at fs=300kHz: 56 dBμV QP.

Reference : Conducted EMI Characteristics And Mitigation Technique

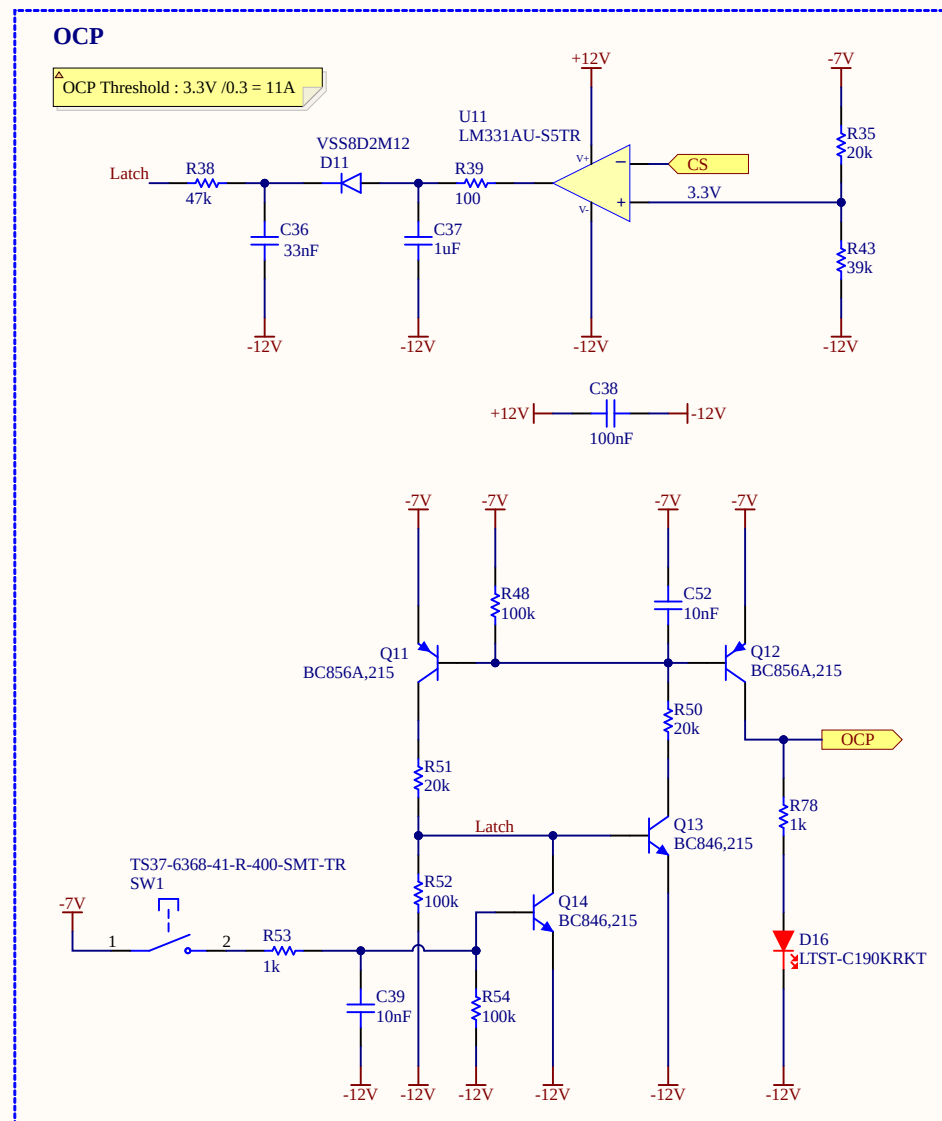
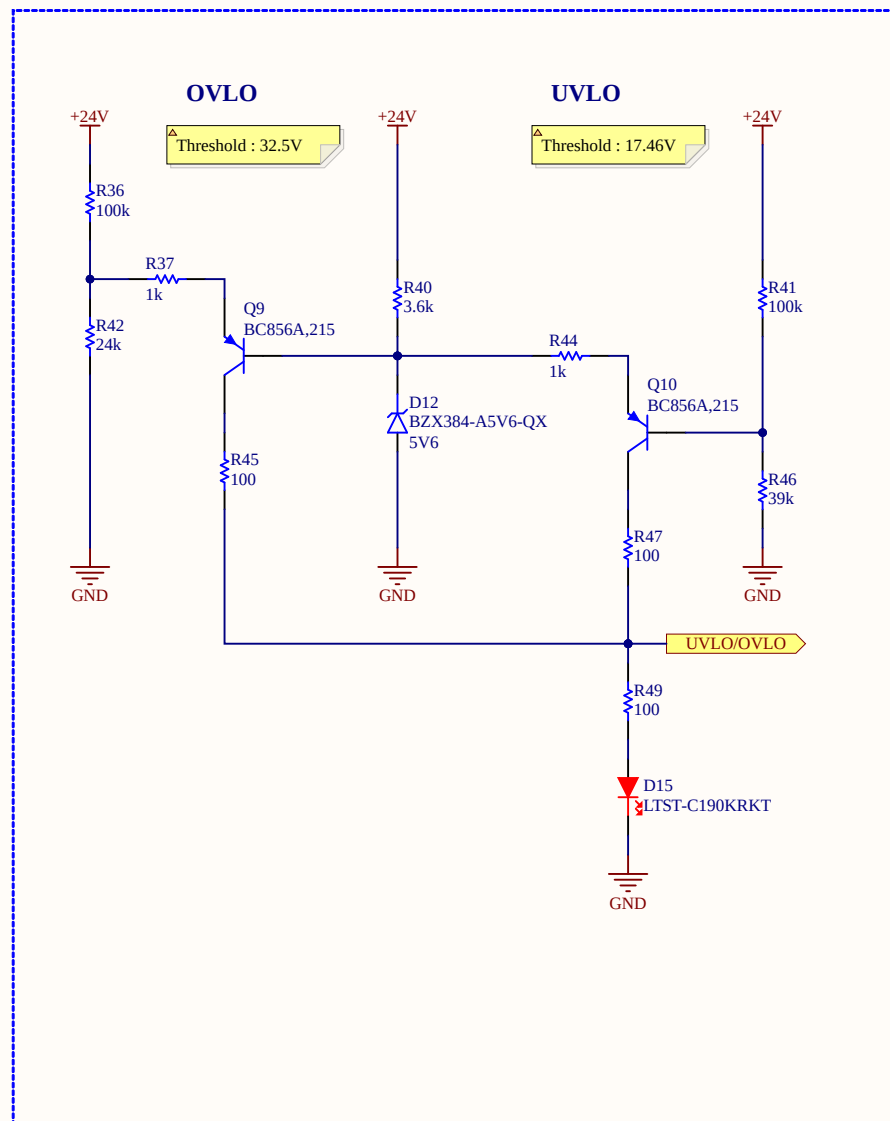
<https://www.ti.com/lit/an/snva489c/snva489c.pdf>

Input filter



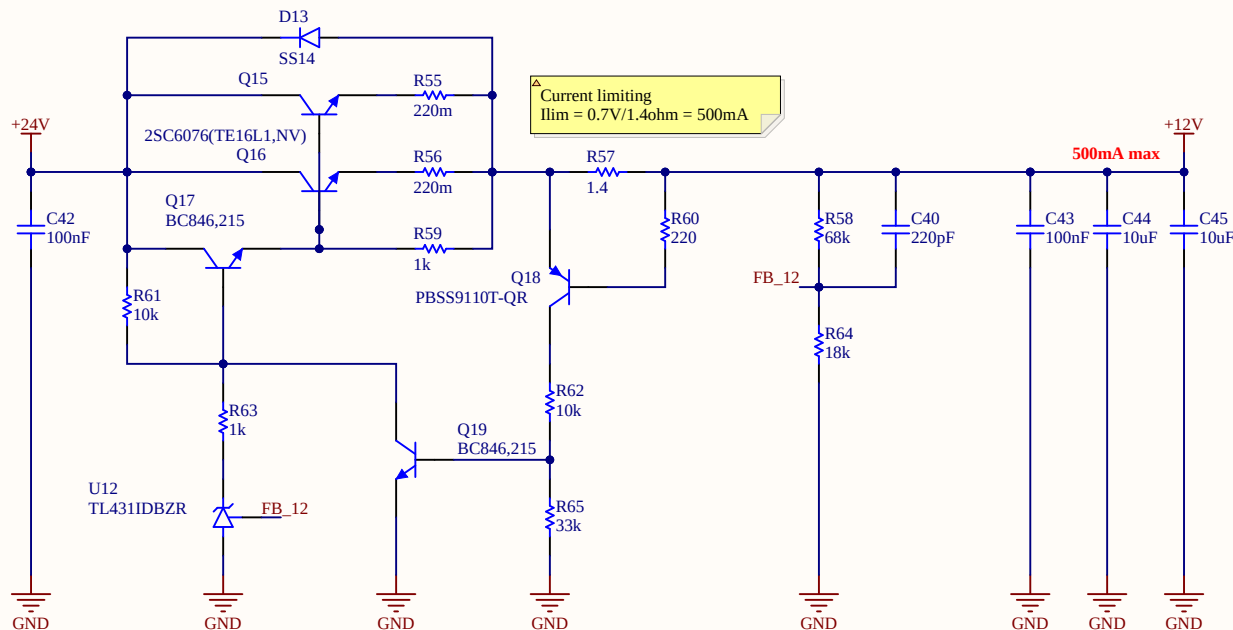
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Protections



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Protections		
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Linear regulators



To obtain the maximum possible current gain using Darlington pair it's important to ensure that the driver transistor has enough collector current to ensure acceptable h_{FE} . When run at very low collector current, most transistors will have a gain that's well below the quoted figure in the datasheet. The base-emitter resistor (shown in the 12V regulator) is sized to ensure that the drive transistor has a collector current that's above the minimum. The resistor also helps to ensure a faster turn-off and minimises leakage current.

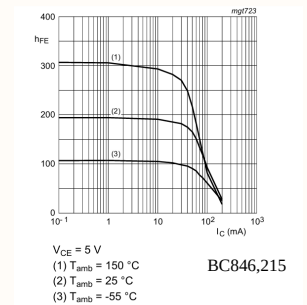
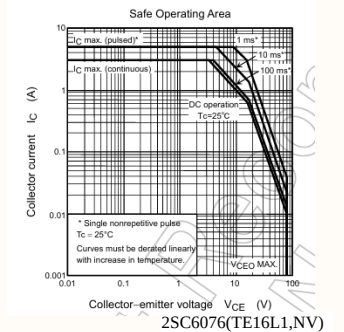
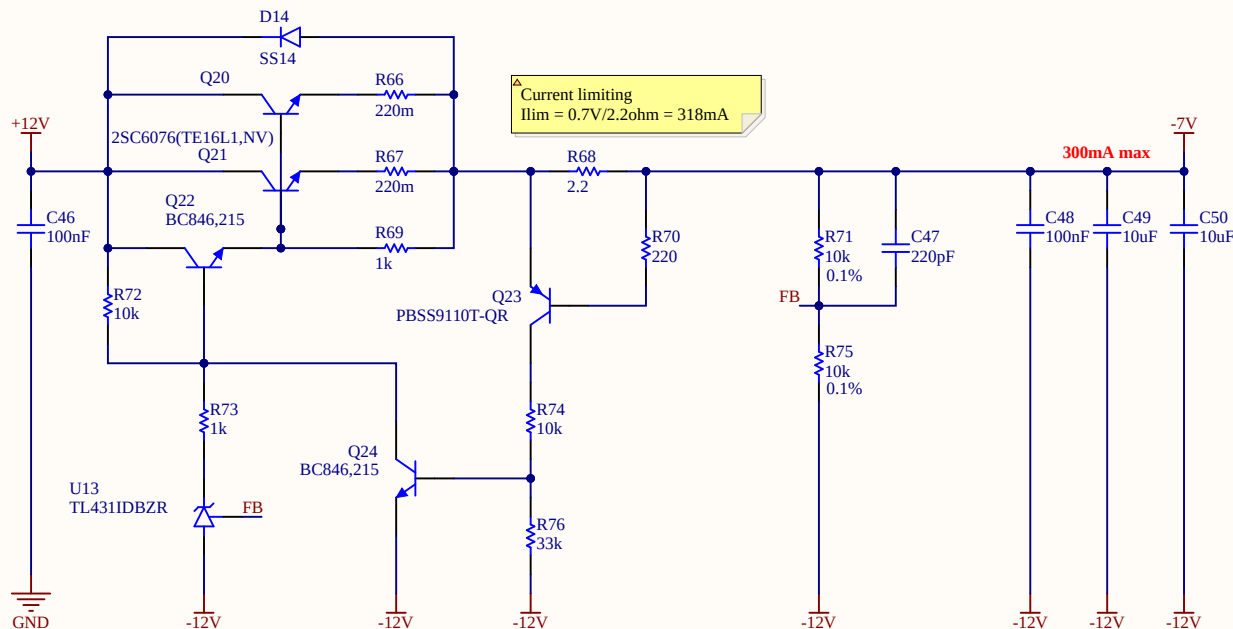


fig. 2. Group A: DC current gain as a function of collector current; typical values



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